Faculty of Computing
Sri Lanka Institute of Information Technology



LAB 08

PC

R.M.S.L Rajapaksha – IT24103715

```
Note that this compiler was either specified at configure time or in
one of several possible environment variables.
--------------------------------------------------------------------------
[ec2-user@ip-172-31-45-239 ~]$ ls -l message1
ls: cannot access 'message1': No such file or directory
[ec2-user@ip-172-31-45-239 ~]$ mpirun --oversubscribe -np 2 ./message1
--------------------------------------------------------------------------
mpirun was unable to launch the specified application as it could not access
or execute an executable:

Executable: ./message1
Node: ip-172-31-45-239

while attempting to start process rank 0.
--------------------------------------------------------------------------
2 total processes failed to start
[ec2-user@ip-172-31-45-239 ~]$ g++ --version
-bash: g++: command not found
[ec2-user@ip-172-31-45-239 ~]$ sudo dnf install -y gcc-c++
Last metadata expiration check: 1:42:58 ago on Thu Oct  8 05:28:18 2026.
Dependencies resolved.
================================================================================
 Package                 Architecture        Version                  Repository                Size
================================================================================
Installing:
 gcc-c++                 x86_64              11.5.0-5.amzn2023.0.5    amazonlinux               15 M
Installing dependencies:
 libstdc++-devel         x86_64              11.5.0-5.amzn2023.0.5    amazonlinux              2.2 M

Transaction Summary
================================================================================
Install  2 Packages

Total download size: 17 M
Installed size: 55 M
Downloading Packages:
(1/2): libstdc++-devel-11.5.0-5.amzn2023.0.5.x86_64.rpm   29 MB/s | 2.2 MB     00:00
(2/2): gcc-c++-11.5.0-5.amzn2023.0.5.x86_64.rpm           66 MB/s |  15 MB     00:00
--------------------------------------------------------------------------------
Total                                                     61 MB/s |  17 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Installing       : libstdc++-devel-11.5.0-5.amzn2023.0.5.x86_64           1/2
  Installing       : gcc-c++-11.5.0-5.amzn2023.0.5.x86_64                   2/2
  Running scriptlet: gcc-c++-11.5.0-5.amzn2023.0.5.x86_64                   2/2
  Verifying        : gcc-c++-11.5.0-5.amzn2023.0.5.x86_64                   1/2
  Verifying        : libstdc++-devel-11.5.0-5.amzn2023.0.5.x86_64           2/2

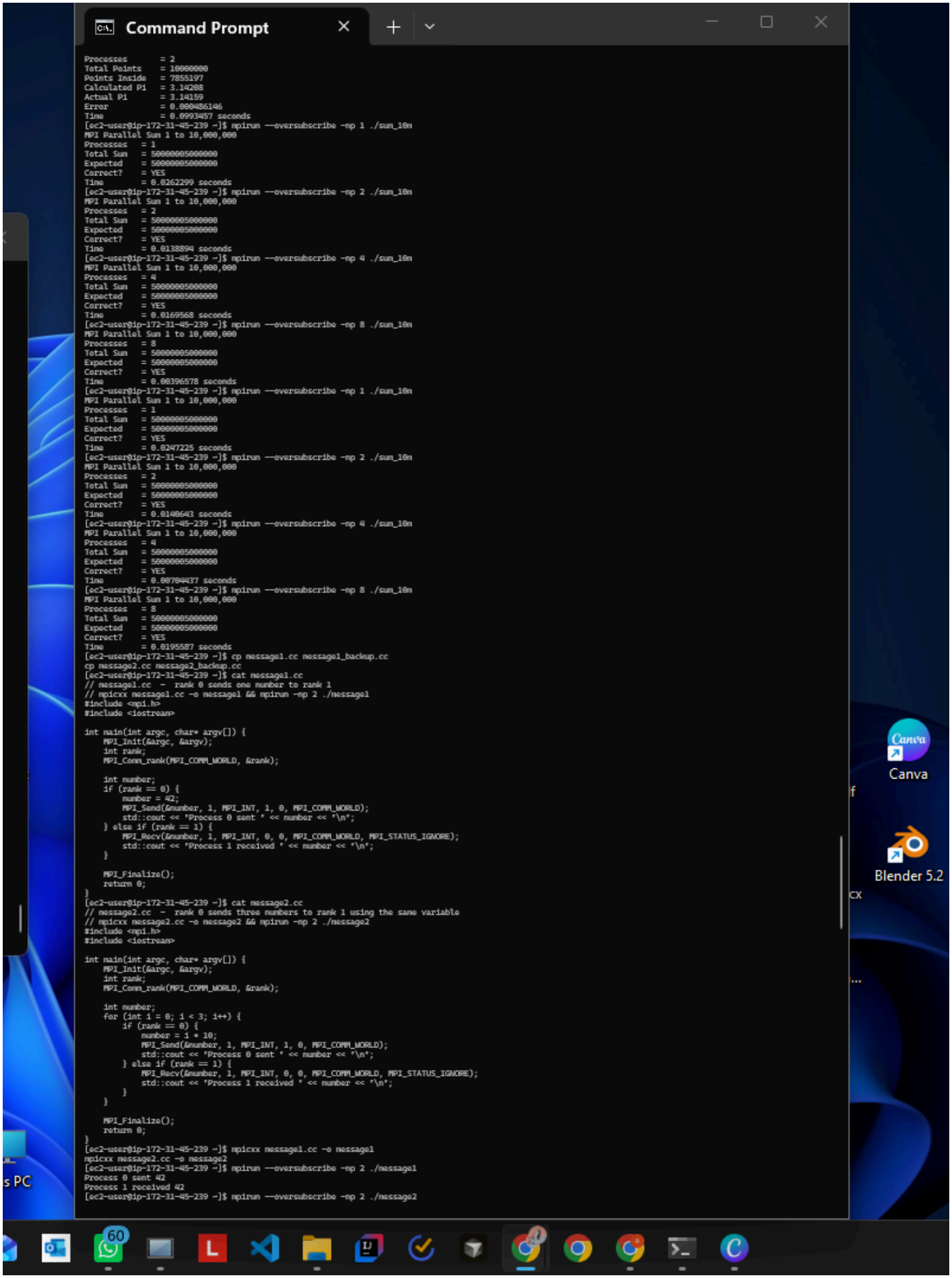
Installed:
  gcc-c++-11.5.0-5.amzn2023.0.5.x86_64
                                        libstdc++-devel-11.5.0-5.amzn2023.0.5.x86_64

Complete!
[ec2-user@ip-172-31-45-239 ~]$ g++ --version
g++ (GCC) 11.5.0 20240719 (Red Hat 11.5.0-5)
Copyright (C) 2021 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

[ec2-user@ip-172-31-45-239 ~]$ export PATH=/usr/lib64/openmpi/bin:$PATH
mpicxx message1.cc -o message1
[ec2-user@ip-172-31-45-239 ~]$ ls -l message1
-rwxr-xr-x. 1 ec2-user ec2-user 90488 Oct  8 07:11 message1
[ec2-user@ip-172-31-45-239 ~]$ mpirun --oversubscribe -np 2 ./message1
Process 0 sent 42
Process 1 received 42
[ec2-user@ip-172-31-45-239 ~]$ mpirun --oversubscribe -np 2 ./message2
--------------------------------------------------------------------------
mpirun was unable to launch the specified application as it could not access
or execute an executable:

Executable: ./message2
Node: ip-172-31-45-239

while attempting to start process rank 0.
--------------------------------------------------------------------------
2 total processes failed to start
[ec2-user@ip-172-31-45-239 ~]$ mpicxx message2.cc -o message2
[ec2-user@ip-172-31-45-239 ~]$ mpirun --oversubscribe -np 2 ./message2
Process 0 sent 0
Process 0 sent 10
Process 0 sent 20
Process 1 received 0
Process 1 received 10
Process 1 received 20
[ec2-user@ip-172-31-45-239 ~]$ export PATH=/usr/lib64/openmpi/bin:$PATH
mpicxx sum_10m.cpp -o sum_10m
[ec2-user@ip-172-31-45-239 ~]$ mpirun --oversubscribe -np 2 ./sum_10m
MPI Parallel Sum 1 to 10,000,000
Processes   = 2
Total Sum   = 50000005000000
Expected    = 50000005000000
Correct?    = YES
Time        = 0.014135 seconds
[ec2-user@ip-172-31-45-239 ~]$ export PATH=/usr/lib64/openmpi/bin:$PATH
mpicxx pi_montecarlo.cpp -o pi_montecarlo
[ec2-user@ip-172-31-45-239 ~]$ mpirun --oversubscribe -np 2 ./pi_montecarlo
MPI Monte Carlo Pi Calculation
Processes       = 2
Total Points    = 10000000
Points Inside   = 7855197
Calculated Pi   = 3.14208
Actual Pi       = 3.14159
Error           = 0.000486146
Time            = 0.0993457 seconds
[ec2-user@ip-172-31-45-239 ~]$ mpirun --oversubscribe -np 1 ./sum_10m
MPI Parallel Sum 1 to 10,000,000
Processes   = 1
Total Sum   = 50000005000000
Expected    = 50000005000000
Correct?    = YES
```


Command Prompt
Processes = 2
Total Points = 10000000
Points Inside = 7855197
Calculated PI = 3.142388
Actual PI = 3.14159
Error = 0.000486146
Time = 0.09934457 seconds
[ec2-user@ip-172-31-45-239 ~]$ mpirun --oversubscribe -np 1 ./sum_10m
MPI Parallel Sum 1 to 10,000,000
Processes = 1
Total Sum = 50000005000000
Expected = 50000005000000
Correct? = YES
Time = 0.0262299 seconds
[ec2-user@ip-172-31-45-239 ~]$ mpirun --oversubscribe -np 2 ./sum_10m
MPI Parallel Sum 1 to 10,000,000
Processes = 2
Total Sum = 50000005000000
Expected = 50000005000000
Correct? = YES
Time = 0.0138894 seconds
[ec2-user@ip-172-31-45-239 ~]$ mpirun --oversubscribe -np 4 ./sum_10m
MPI Parallel Sum 1 to 10,000,000
Processes = 4
Total Sum = 50000005000000
Expected = 50000005000000
Correct? = YES
Time = 0.0169568 seconds
[ec2-user@ip-172-31-45-239 ~]$ mpirun --oversubscribe -np 8 ./sum_10m
MPI Parallel Sum 1 to 10,000,000
Processes = 8
Total Sum = 50000005000000
Expected = 50000005000000
Correct? = YES
Time = 0.08396578 seconds
[ec2-user@ip-172-31-45-239 ~]$ mpirun --oversubscribe -np 1 ./sum_10m
MPI Parallel Sum 1 to 10,000,000
Processes = 1
Total Sum = 50000005000000
Expected = 50000005000000
Correct? = YES
Time = 0.0247225 seconds
[ec2-user@ip-172-31-45-239 ~]$ mpirun --oversubscribe -np 2 ./sum_10m
MPI Parallel Sum 1 to 10,000,000
Processes = 2
Total Sum = 50000005000000
Expected = 50000005000000
Correct? = YES
Time = 0.0140643 seconds
[ec2-user@ip-172-31-45-239 ~]$ mpirun --oversubscribe -np 4 ./sum_10m
MPI Parallel Sum 1 to 10,000,000
Processes = 4
Total Sum = 50000005000000
Expected = 50000005000000
Correct? = YES
Time = 0.087044437 seconds
[ec2-user@ip-172-31-45-239 ~]$ mpirun --oversubscribe -np 8 ./sum_10m
MPI Parallel Sum 1 to 10,000,000
Processes = 8
Total Sum = 50000005000000
Expected = 50000005000000
Correct? = YES
Time = 0.0195587 seconds
[ec2-user@ip-172-31-45-239 ~]$ cp message1.cc message1_backup.cc
cp message2.cc message2_backup.cc
[ec2-user@ip-172-31-45-239 ~]$ cat message1.cc
// message1.cc - rank 0 sends one number to rank 1
// mpicxx message1.cc -o message1 && mpirun -np 2 ./message1
#include <mpi.h>
#include <iostream>
int main(int argc, char* argv[]) {
MPI_Init(&argc, &argv);
int rank;
MPI_Comm_rank(MPI_COMM_WORLD, &rank);
int number;
if (rank == 0) {
number = 42;
MPI_Send(&number, 1, MPI_INT, 1, 0, MPI_COMM_WORLD);
std::cout << "Process 0 sent " << number << "\n";
} else if (rank == 1) {
MPI_Recv(&number, 1, MPI_INT, 0, 0, MPI_COMM_WORLD, MPI_STATUS_IGNORE);
std::cout << "Process 1 received " << number << "\n";
}
MPI_Finalize();
return 0;
}
[ec2-user@ip-172-31-45-239 ~]$ cat message2.cc
// message2.cc - rank 0 sends three numbers to rank 1 using the same variable
// mpicxx message2.cc -o message2 && mpirun -np 2 ./message2
#include <mpi.h>
#include <iostream>
int main(int argc, char* argv[]) {
MPI_Init(&argc, &argv);
int rank;
MPI_Comm_rank(MPI_COMM_WORLD, &rank);
int number;
for (int i = 0; i < 3; i++) {
if (rank == 0) {
number = i * 10;
MPI_Send(&number, 1, MPI_INT, 1, 0, MPI_COMM_WORLD);
std::cout << "Process 0 sent " << number << "\n";
} else if (rank == 1) {
MPI_Recv(&number, 1, MPI_INT, 0, 0, MPI_COMM_WORLD, MPI_STATUS_IGNORE);
std::cout << "Process 1 received " << number << "\n";
}
}
MPI_Finalize();
return 0;
}
[ec2-user@ip-172-31-45-239 ~]$ mpicxx message1.cc -o message1
mpicxx message2.cc -o message2
[ec2-user@ip-172-31-45-239 ~]$ mpirun --oversubscribe -np 2 ./message1
Process 0 sent 42
Process 1 received 42
[ec2-user@ip-172-31-45-239 ~]$ mpirun --oversubscribe -np 2 ./message2
Canva
Blender 5.2

Command Prompt

```
    MPI_Finalize();
    return 0;
}
[ec2-user@ip-172-31-45-239 ~]$ mpicxx message1.cc -o message1
mpicxx message2.cc -o message2
[ec2-user@ip-172-31-45-239 ~]$ mpirun --oversubscribe -np 2 ./message1
Process 0 sent 42
Process 1 received 42
[ec2-user@ip-172-31-45-239 ~]$ mpirun --oversubscribe -np 2 ./message2
Process 0 sent 0
Process 0 sent 10
Process 0 sent 20
Process 1 received 0
Process 1 received 10
Process 1 received 20
[ec2-user@ip-172-31-45-239 ~]$ sed -i 's/MPI_Recv(&number, 1, MPI_INT, 0,/MPI_Recv(\&number, 1, MPI_INT, 1,/' message2.cc
[ec2-user@ip-172-31-45-239 ~]$ grep MPI_Recv message2.cc
            MPI_Recv(&number, 1, MPI_INT, 1, 0, MPI_COMM_WORLD, MPI_STATUS_IGNORE);
[ec2-user@ip-172-31-45-239 ~]$ mpicxx message2.cc -o message2
[ec2-user@ip-172-31-45-239 ~]$ mpirun --oversubscribe -np 2 ./message2
Process 0 sent 0
Process 0 sent 10
Process 0 sent 20
^C[ec2-user@ip-172-31-45-239 ~]$
[ec2-user@ip-172-31-45-239 ~]$ mpirun --oversubscribe -np 2 ./message2
Process 0 sent 0
Process 0 sent 10
Process 0 sent 20
cp message2_backup.cc message2.cc
^C
[ec2-user@ip-172-31-45-239 ~]$ cp message2_backup.cc message2.cc
[ec2-user@ip-172-31-45-239 ~]$ grep MPI_Send message2.cc
            MPI_Send(&number, 1, MPI_INT, 1, 0, MPI_COMM_WORLD);
[ec2-user@ip-172-31-45-239 ~]$ cat > message2.cc <<'EOF'
#include <mpi.h>
#include <iostream>

using namespace std;

int main(int argc, char* argv[])
{
    MPI_Init(&argc, &argv);

    int rank;
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);

    int number;

    // Buffer for MPI_Bsend
    int buffer_size = 3 * (MPI_BSEND_OVERHEAD + sizeof(int));
    void* buffer = malloc(buffer_size);

    MPI_Buffer_attach(buffer, buffer_size);

    for (int i = 0; i < 3; i++)
    {
        if (rank == 0)
        {
            number = i * 10;

            MPI_Bsend(
                &number,
                1,
                MPI_INT,
                1,
                0,
                MPI_COMM_WORLD
            );

            cout << "Process 0 sent " << number << "\n";
        }
        else if (rank == 1)
        {
            MPI_Recv(
                &number,
                1,
                MPI_INT,
                0,
                0,
                MPI_COMM_WORLD,
                MPI_STATUS_IGNORE
            );

            cout << "Process 1 received " << number << "\n";
        }
    }

    MPI_Buffer_detach(&buffer, &buffer_size);
    free(buffer);

    MPI_Finalize();

    return 0;
}
EOF
[ec2-user@ip-172-31-45-239 ~]$ mpicxx message2.cc -o message2
[ec2-user@ip-172-31-45-239 ~]$ mpirun --oversubscribe -np 2 ./message2
Process 0 sent 0
Process 1 received 0
Process 0 sent 10
Process 0 sent 20
Process 1 received 10
Process 1 received 20
[ec2-user@ip-172-31-45-239 ~]$ cat > pi_any_source.cpp <<'EOF'
#include <mpi.h>
#include <iostream>
#include <cstdlib>
#include <ctime>
#include <cmath>

using namespace std;

int main(int argc, char** argv)
{
    MPI_Init(&argc, &argv);

    int rank, size;
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    const long long TOTAL_POINTS = 10000000LL;
```

Command Prompt

```
    static_cast<unsigned int>(time(NULL)) + rank * 12345;

long long local_inside = 0;

for (long long i = 0; i < points_per_process; i++)
{
    double x = (double)rand_r(&seed) / RAND_MAX;
    double y = (double)rand_r(&seed) / RAND_MAX;

    if (x * x + y * y <= 1.0)
        local_inside++;
}

if (rank != 0)
{
    MPI_Send(
        &local_inside,
        1,
        MPI_LONG_LONG,
        0,
        0,
        MPI_COMM_WORLD
    );
}
else
{
    long long total_inside = local_inside;

    for (int i = 1; i < size; i++)
    {
        long long received;

        MPI_Recv(
            &received,
            1,
            MPI_LONG_LONG,
            MPI_ANY_SOURCE,
            0,
            MPI_COMM_WORLD,
            MPI_STATUS_IGNORE
        );

        total_inside += received;
    }

    double pi = 4.0 * total_inside / TOTAL_POINTS;

    cout << "MPI Monte Carlo Pi - MPI_ANY_SOURCE" << endl;
    cout << "Processes     = " << size << endl;
    cout << "Total Points  = " << TOTAL_POINTS << endl;
    cout << "Calculated Pi = " << pi << endl;
    cout << "Actual Pi     = " << M_PI << endl;
    cout << "Error         = " << fabs(M_PI - pi) << endl;
}

MPI_Finalize();

return 0;
}
EOF
[ec2-user@ip-172-31-45-239 ~]$ mpicxx pi_any_source.cpp -o pi_any_source
[ec2-user@ip-172-31-45-239 ~]$ mpirun --oversubscribe -np 2 ./pi_any_source
MPI Monte Carlo Pi - MPI_ANY_SOURCE
Processes     = 2
Total Points  = 10000000
Calculated Pi = 3.14218
Actual Pi     = 3.14159
Error         = 0.000582546
[ec2-user@ip-172-31-45-239 ~]$ cat > pi_any_source_bsend.cpp <<'EOF'
#include <mpi.h>
#include <iostream>
#include <cstdlib>
#include <ctime>
#include <cmath>

using namespace std;

int main(int argc, char** argv)
{
    MPI_Init(&argc, &argv);

    int rank, size;
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    const long long TOTAL_POINTS = 10000000LL;

    long long points_per_process = TOTAL_POINTS / size;

    if (rank == size - 1)
        points_per_process += TOTAL_POINTS % size;

    unsigned int seed =
        static_cast<unsigned int>(time(NULL)) + rank * 12345;

    long long local_inside = 0;

    for (long long i = 0; i < points_per_process; i++)
    {
        double x = (double)rand_r(&seed) / RAND_MAX;
        double y = (double)rand_r(&seed) / RAND_MAX;

        if (x * x + y * y <= 1.0)
            local_inside++;
    }

    if (rank != 0)
    {
        int buffer_size = MPI_BSEND_OVERHEAD + sizeof(long long);
        void* buffer = malloc(buffer_size);

        MPI_Buffer_attach(buffer, buffer_size);

        MPI_Bsend(
            &local_inside,
            1,
            MPI_LONG_LONG,
            0,
            0,
            MPI_COMM_WORLD
        );
```

```
        double y = (double)rand_r(&seed) / RAND_MAX;

        if (x * x + y * y <= 1.0)
            local_inside++;
    }

    if (rank != 0)
    {
        int buffer_size = MPI_BSEND_OVERHEAD + sizeof(long long);
        void* buffer = malloc(buffer_size);

        MPI_Buffer_attach(buffer, buffer_size);

        MPI_Bsend(
            &local_inside,
            1,
            MPI_LONG_LONG,
            0,
            0,
            MPI_COMM_WORLD
        );

        MPI_Buffer_detach(&buffer, &buffer_size);
        free(buffer);
    }
    else
    {
        long long total_inside = local_inside;

        for (int i = 1; i < size; i++)
        {
            long long received;

            MPI_Recv(
                &received,
                1,
                MPI_LONG_LONG,
                MPI_ANY_SOURCE,
                0,
                MPI_COMM_WORLD,
                MPI_STATUS_IGNORE
            );

            total_inside += received;
        }

        double pi = 4.0 * total_inside / TOTAL_POINTS;

        cout << "MPI Monte Carlo Pi - BSend + MPI_ANY_SOURCE" << endl;
        cout << "Processes     = " << size << endl;
        cout << "Total Points  = " << TOTAL_POINTS << endl;
        cout << "Calculated Pi = " << pi << endl;
        cout << "Actual Pi     = " << M_PI << endl;
        cout << "Error         = " << fabs(M_PI - pi) << endl;
    }

    MPI_Finalize();

    return 0;
}
EOF
[ec2-user@ip-172-31-45-239 ~]$ mpicxx pi_any_source_bsend.cpp -o pi_any_source_bsend
[ec2-user@ip-172-31-45-239 ~]$ mpirun --oversubscribe -np 2 ./pi_any_source_bsend
MPI Monte Carlo Pi - BSend + MPI_ANY_SOURCE
Processes     = 2
Total Points  = 10000000
Calculated Pi = 3.14181
Actual Pi     = 3.14159
Error         = 0.0002216946
[ec2-user@ip-172-31-45-239 ~]$ cat message1.cc
// message1.cc  -  rank 0 sends one number to rank 1
// mpicxx message1.cc -o message1 && mpirun -np 2 ./message1
#include <mpi.h>
#include <iostream>

int main(int argc, char* argv[]) {
    MPI_Init(&argc, &argv);
    int rank;
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);

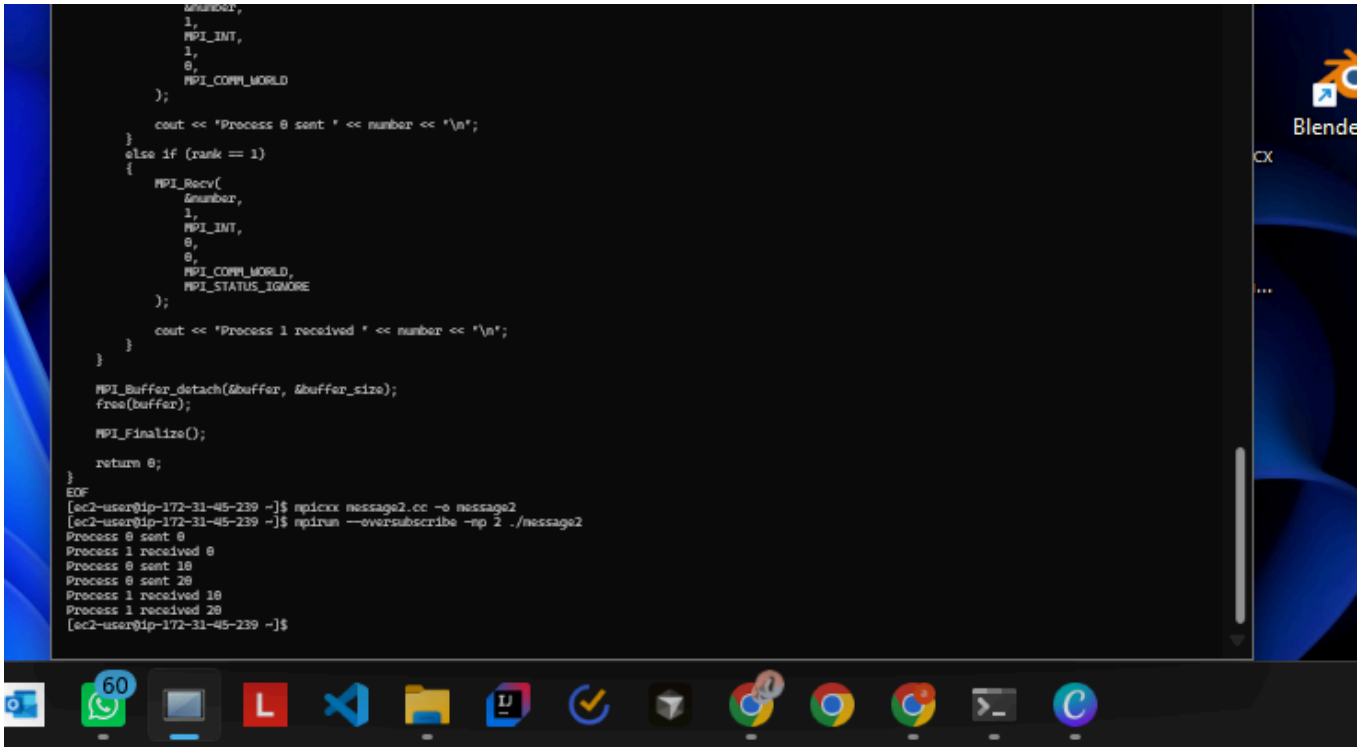
    int number;
    if (rank == 0) {
        number = 42;
        MPI_Send(&number, 1, MPI_INT, 1, 0, MPI_COMM_WORLD);
        std::cout << "Process 0 sent " << number << "\n";
    } else if (rank == 1) {
        MPI_Recv(&number, 1, MPI_INT, 0, 0, MPI_COMM_WORLD, MPI_STATUS_IGNORE);
        std::cout << "Process 1 received " << number << "\n";
    }

    MPI_Finalize();
    return 0;
}
[ec2-user@ip-172-31-45-239 ~]$ Connection to 52.73.199.20 closed by remote host.
Connection to 52.73.199.20 closed.

E:\Git Hub Projects\PC AWS Projects\LAB 07>cat message1.cc
'cat' is not recognized as an internal or external command,
operable program or batch file.

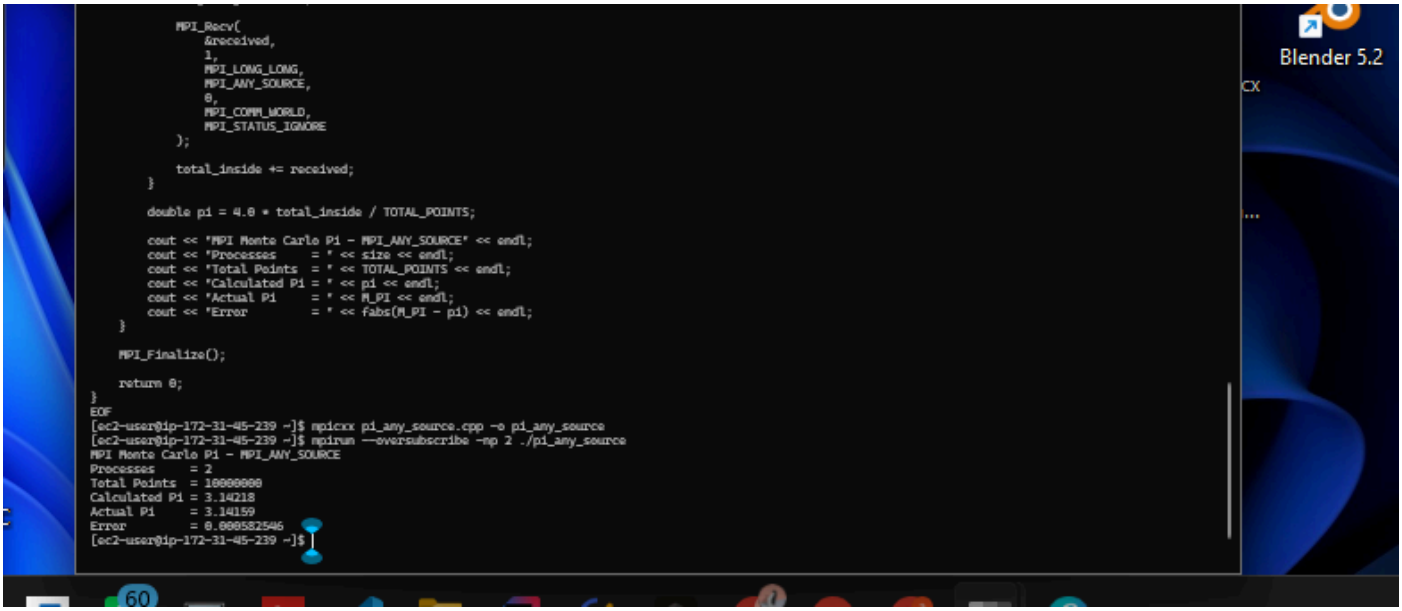
E:\Git Hub Projects\PC AWS Projects\LAB 07>
```

Exercise 5 – Buffered Send (MPI_Bsend)



Exercise 6 – MPI_ANY_SOURCE

The receiving process uses MPI_ANY_SOURCE to accept messages from any MPI process instead of specifying a particular source process.



Exercise 7 – Buffered Send with MPI_ANY_SOURCE

The Monte Carlo Pi calculation uses MPI_Bsend for buffered communication and MPI_ANY_SOURCE to receive messages from any source process.

